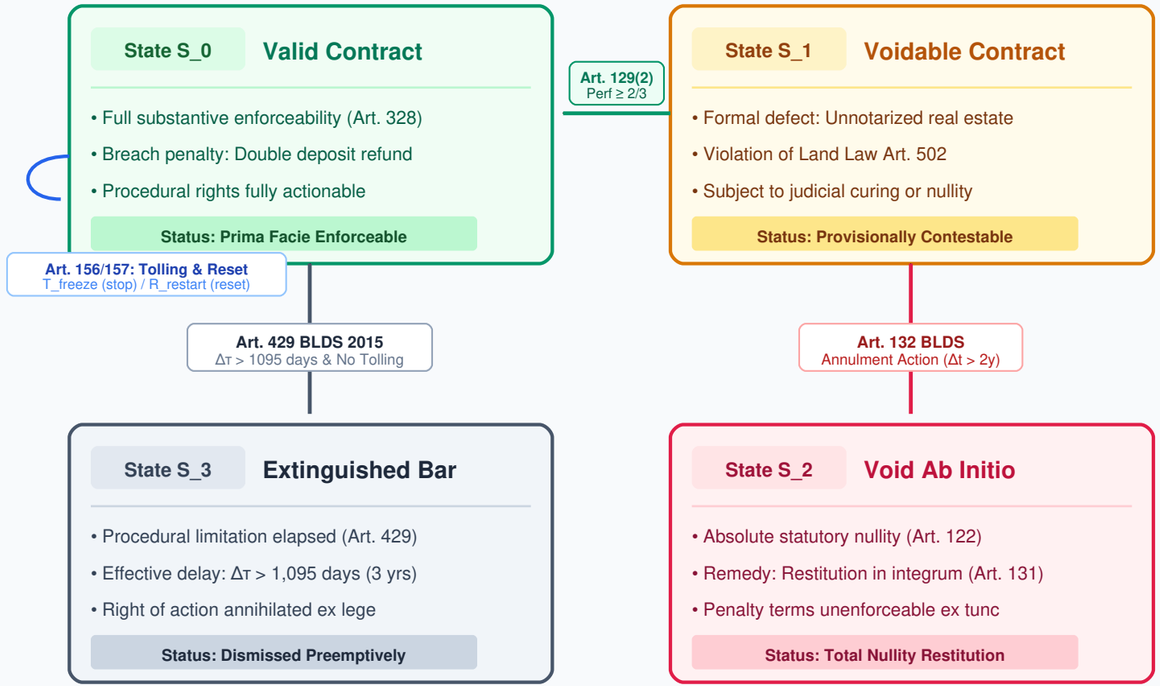
(a) 4-State Civil Procedural Automaton M_state = ⟨S, Σ, δ, s_0, F⟩

Non-Monotonic Contract State-Transitions under Civil Code (BLDS 2015)

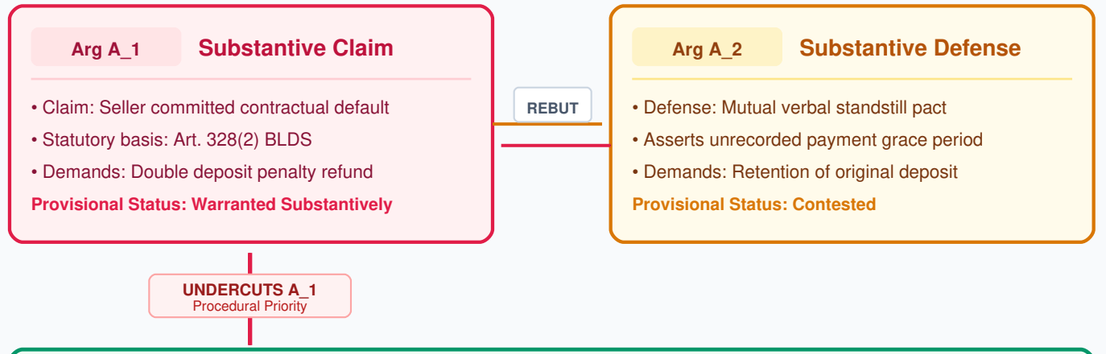


Axiomatic Orthogonality: S_0 ⊓ S_1 ⊑ ⊥, S_0 ⊓ S_3 ⊑ ⊥, S_0 ⊓ S_2 ⊑ ⊥

Markovian transitions prevent logical explosion while preserving non-monotonicity

(b) Dung Defeasible Argumentation Graph AF = ⟨Args, Defeat⟩

Asymmetric Procedural Undercutting & Knaster-Tarski Grounded Extension



Dung Characteristic Operator & Fixed Point Adjudication Closure

$$\Gamma_{AF}(S) = \{ A \in Args \mid \forall B \in Args, (B, A) \in Defeat \Rightarrow \exists C \in S, (C, B) \in Defeat \}$$

Theorem 2: Complete Lattice Knaster-Tarski Monotonic Operator Guarantee

GROUNDED EXTENSION: GE(AF) = lfp(Γ_AF) = { A_3 }

Substantive claims A_1 and A_2 defeated ⟹ Final Adjudicated Decree: DISMISSED AS STATUTE-BARRED